Question 8: Record all Dockerfiles, Kubernetes manifests, commands, screenshots, and observations for each experiment.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. Repository structure

https://github.com/meenakshi16rp/Devops-lab-2

```
ensp461-lab2/
  app/               sample Flask application + unit tests
  Dockerfile         application image (python:3.12-slim)
  docker-compose.yml multi-container setup (web + redis)
  k8s/configmap.yaml k8s/deployment.yaml k8s/service.yaml
  Jenkinsfile-CD     continuous deployment pipeline
  docs/logs/         captured command output for every experiment
  docs/screenshots/  terminal renders and Jenkins screenshots
```

2. Tool versions

recorded before the experiments

```
meenakshi@ubuntu:~$ docker version --format {{.Server.Version}}
29.9.2
meenakshi@ubuntu:~$ docker compose version
Docker Compose version v5.7.0
meenakshi@ubuntu:~$ kubectl version --client
Client Version: v1.37.2
meenakshi@ubuntu:~$ minikube version
minikube version: v1.39.0
meenakshi@ubuntu:~$ minikube kubectl -- kubectl version ... v1.37.0
```

3. Command record per experiment

Experiment	Key commands	Evidence
Q1 Containerize	docker build / run / history / inspect / logs	r2-q1-*.png, docs/logs/q1-*.txt
Q2 Manage	docker ps / stop / start / tag / login / push, docker compose up / down	r2-q2-*.png, docs/logs/q2-*.txt
Q3 Storage	docker exec, rm, volume create, -v mounts	r2-q3-*.png, docs/logs/q3-storage.txt
Q4 Network	docker network create, exec ping / wget	r2-q4-network.png, docs/logs/q4-network.txt
Q5 K8s deploy	minikube start, kubectl create secret, apply, rollout status	r2-q5-*.png, docs/logs/q5-*.txt
Q6 Strategies	kubectl set image, rollout status / history / undo, scale	r2-q6-*.png, docs/logs/q6-*.txt
Q7 CD	git push, Jenkins ensp461-lab2-cd, kubectl apply + smoke test	07-*.png, r2-q7-*.png, docs/logs/q7-*.txt

4. Observations

- Every command was executed as user meenakshi on Ubuntu 24.04 LTS (WSL2); outputs were captured headlessly (tee) and are committed under docs/logs/.
- The Dockerfile, Compose file and manifests are the single source of truth - the same image runs standalone, in Compose and on Kubernetes.

- Nothing in the Docker or Kubernetes state references any other user identity: images are ghcr.io/meenakshi16rp/*, resources are named lab2-*.